

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimeter-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 903/2
Bridge Chainage – 906+860

Project - Chittorgarh to Kota
Zone – RJ 06
Structure Type – MNB

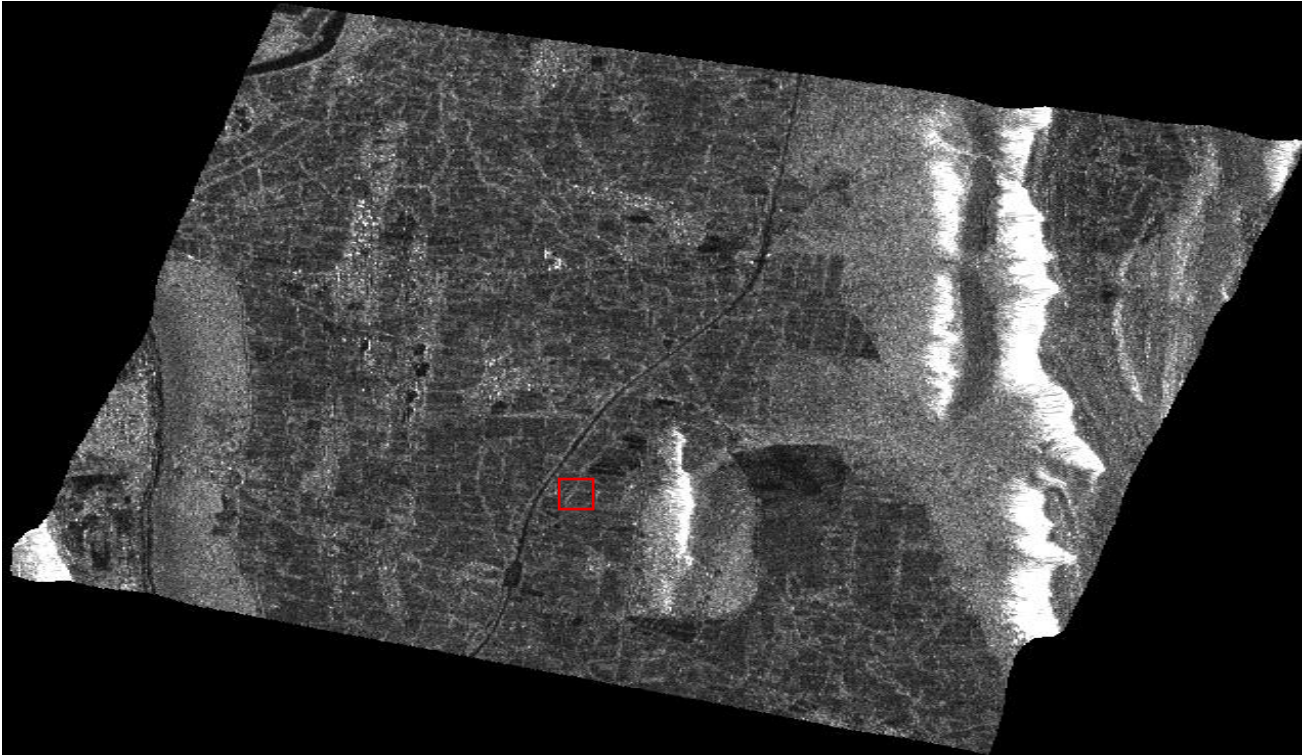
Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

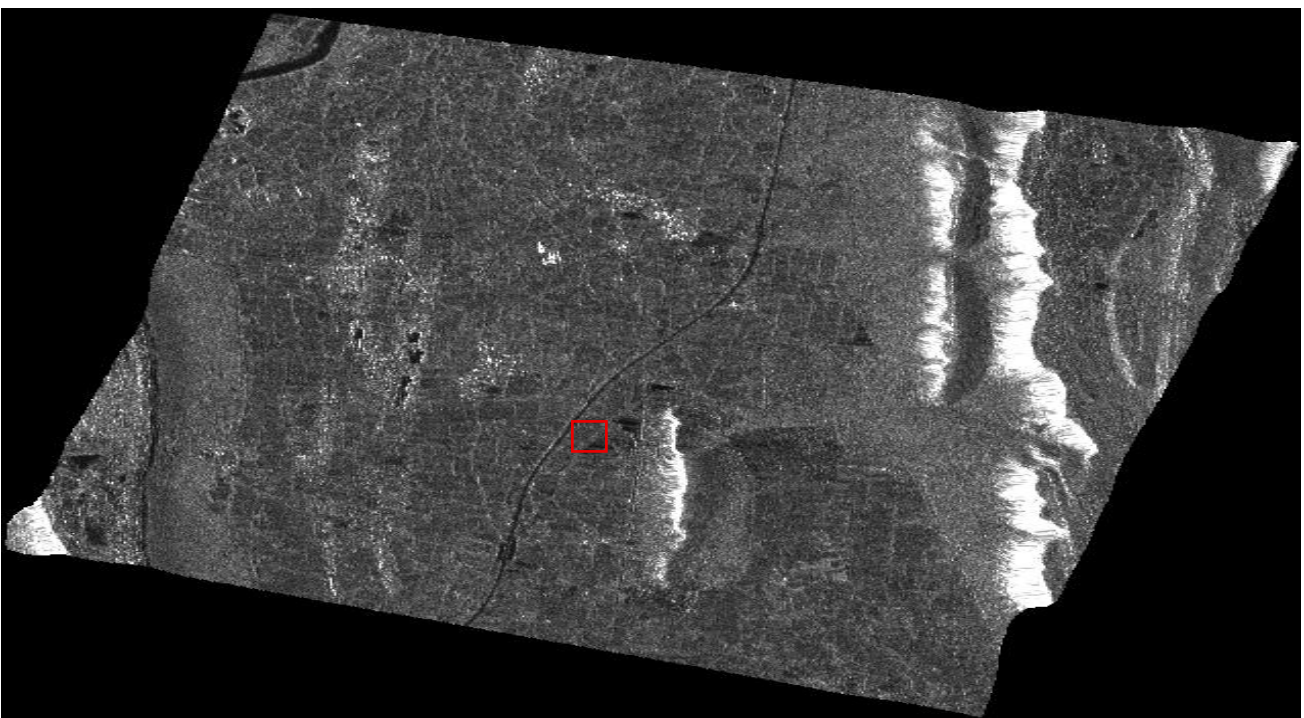
Flight Direction - DESCENDING
Polarization – VV+VH
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

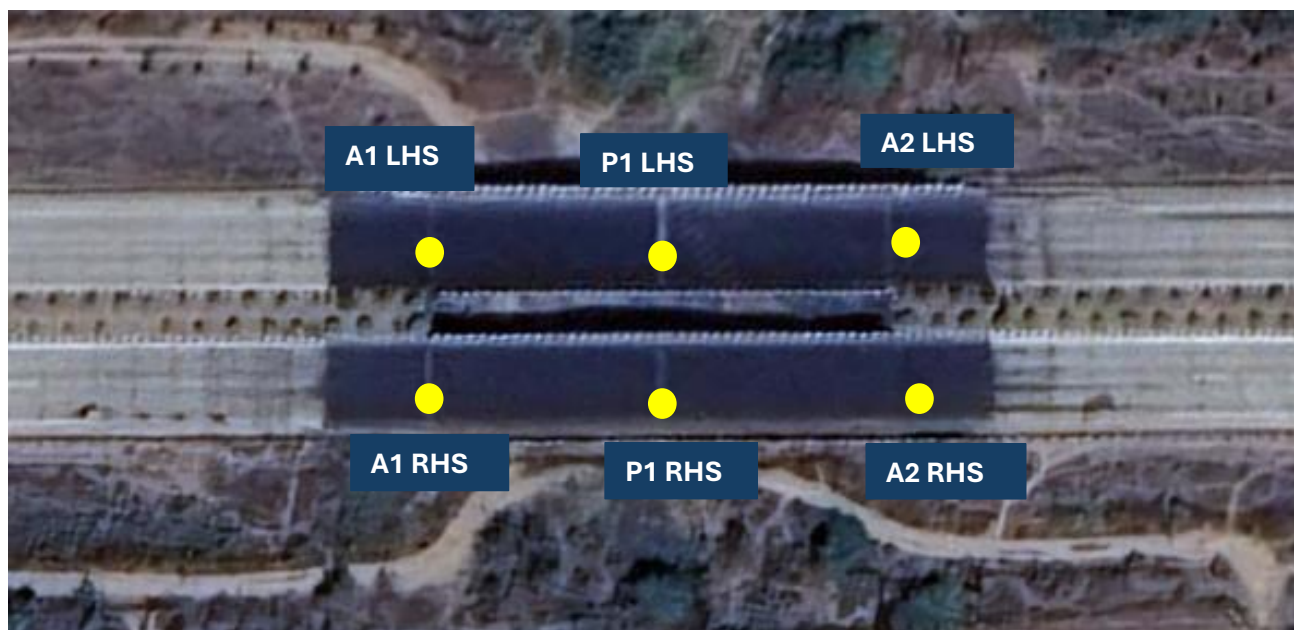


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

MNB- CH-906+860

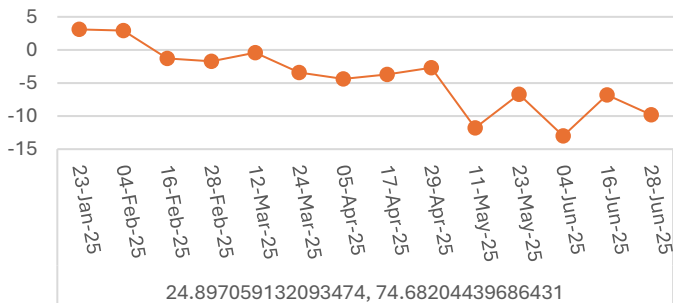
Length – 42.59M



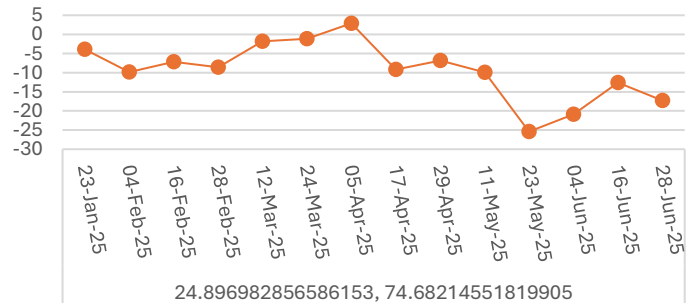
Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
P1	Pier 1		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025

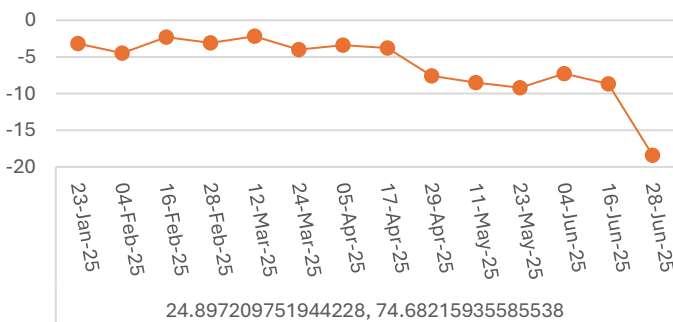
Displacement @A1 LHS VV



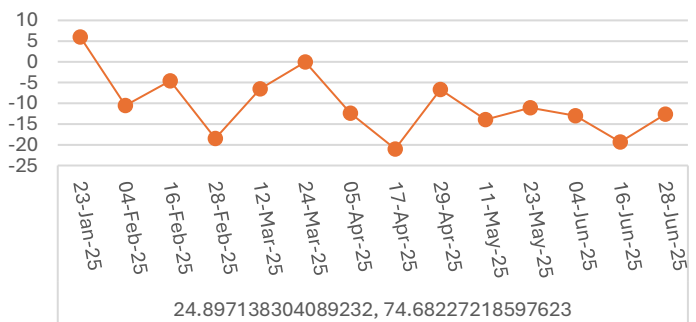
Displacement @A1 RHS VV



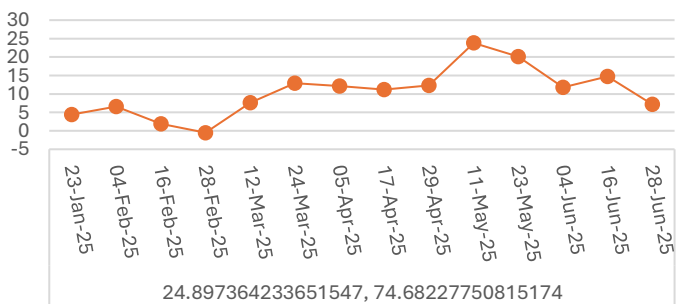
Displacement @P1 LHS VV



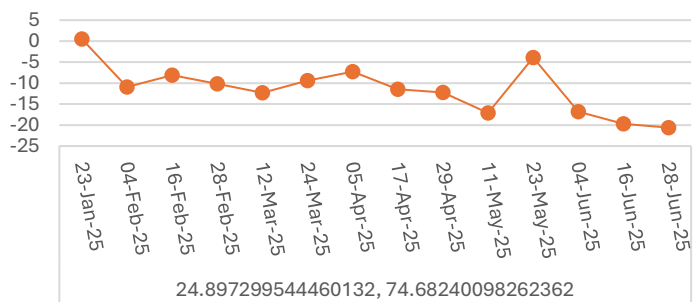
Displacement @P1 RHS VV



Displacement @A2 LHS VV

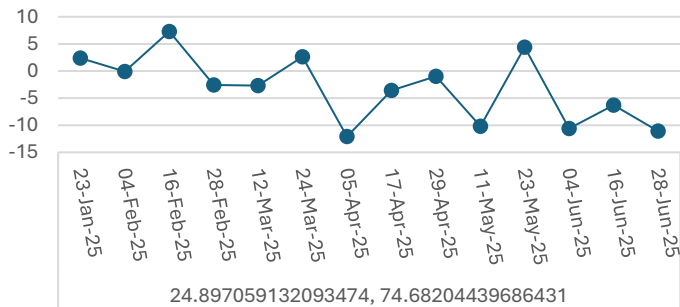


Displacement @A2 RHS VV

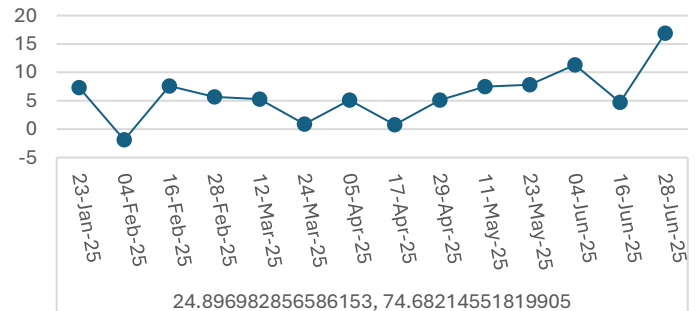


3.2 VH Displacement From 23 Jan 2025 To 28 June 2025

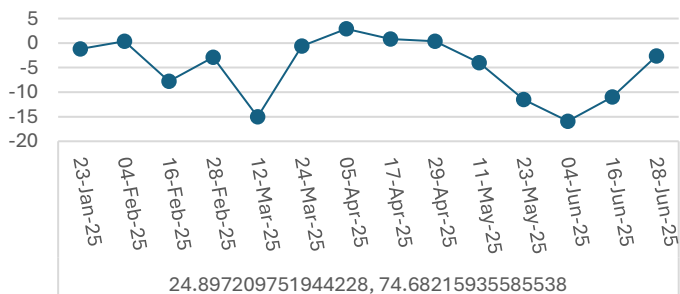
Displacement @A1 LHS VH



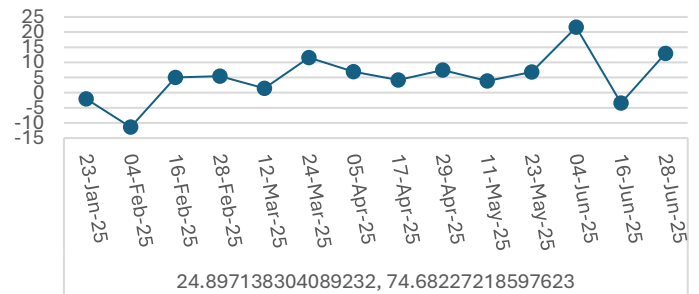
Displacement @A1 RHS VH



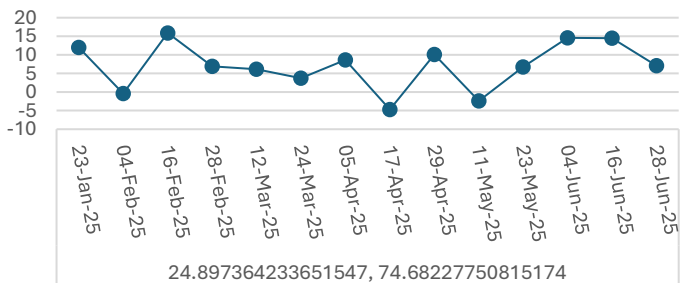
Displacement @P1 LHS VH



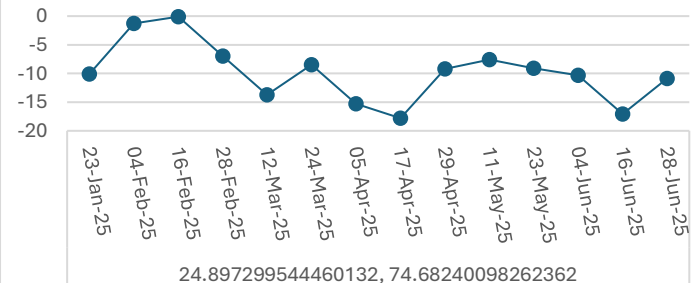
Displacement @P1 RHS VH



Displacement @A2 LHS VH



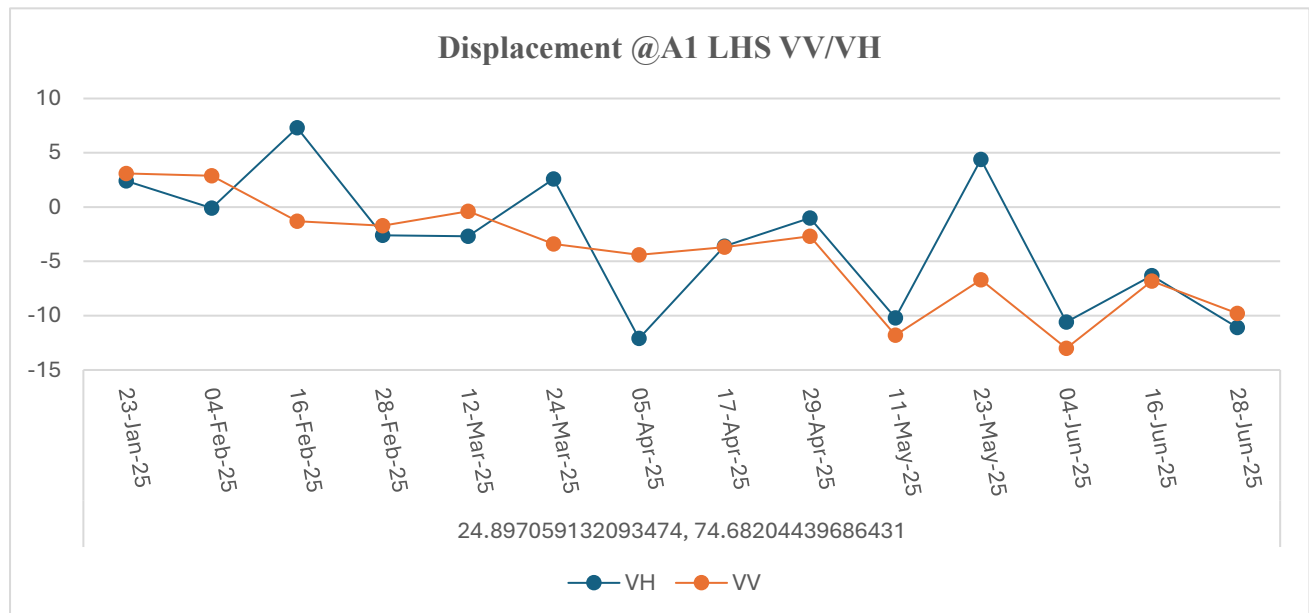
Displacement @A2 RHS VH



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.897059132093474, 74.68204439686431)



Key Observations:

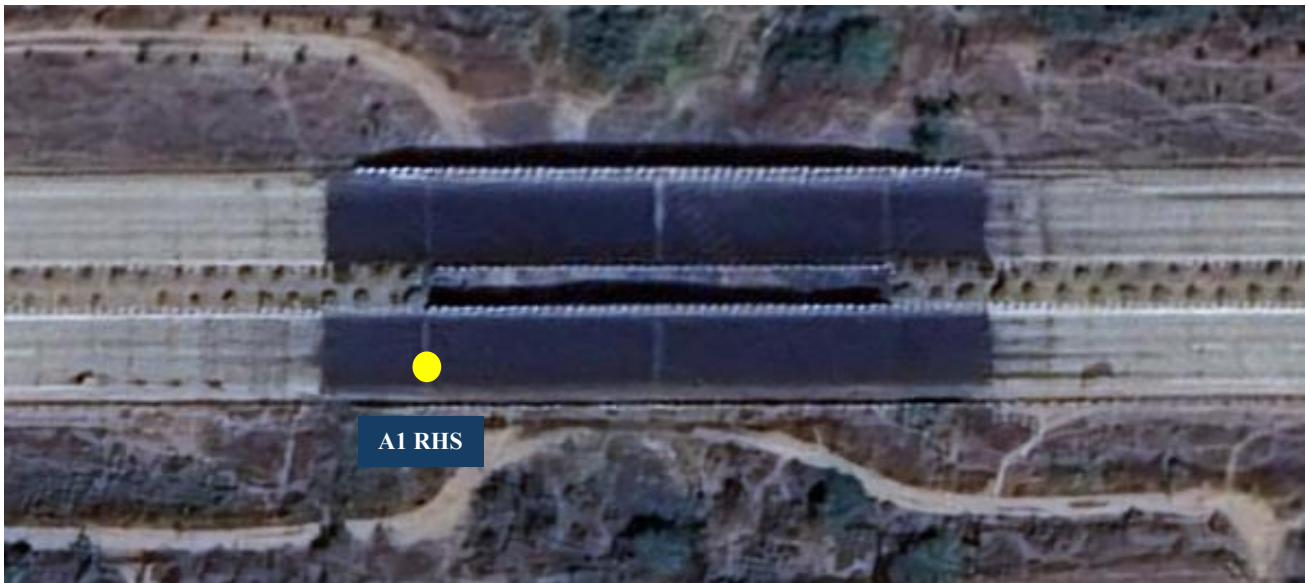
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement exhibits moderate variability with a predominantly negative trend over the monitoring period (Jan–Jun 2025), indicating overall vertical settlement at the A1 LHS location.
- **Horizontal displacement (VH)** Horizontal displacement shows high variability with alternating uplift and settlement phases, ranging roughly between +7 mm and –12 mm.

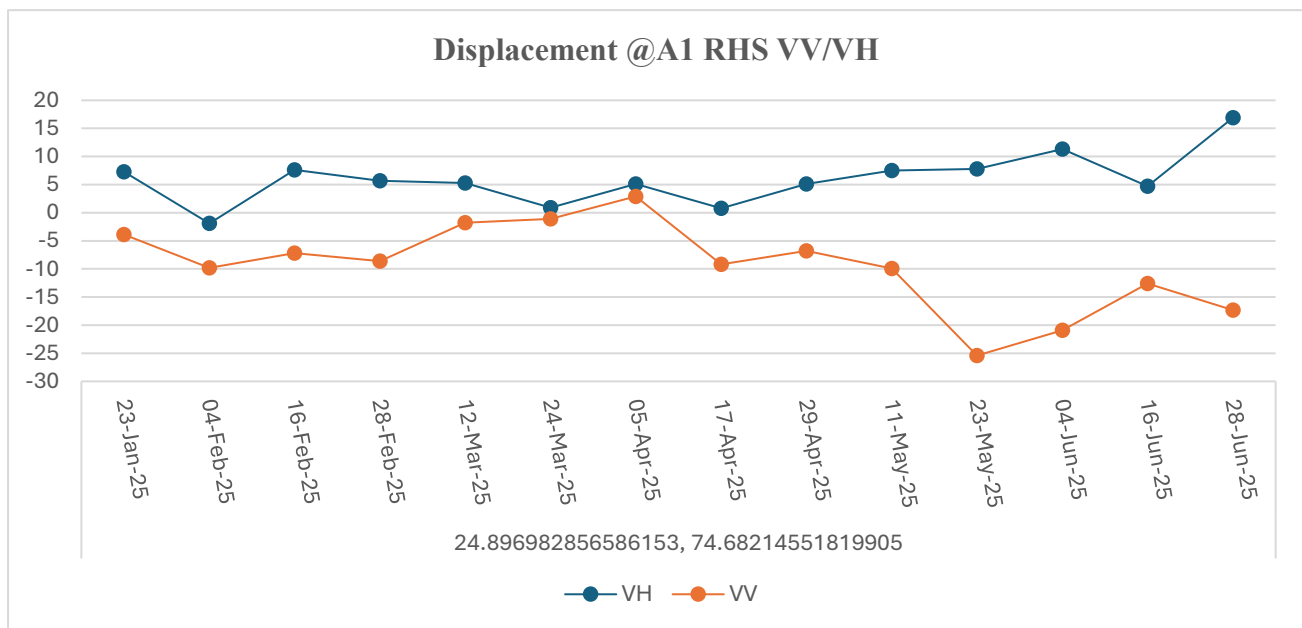
Critical Events & Anomalies:

- **January 2025:** Both VH and VV remain near neutral to slightly positive values, indicating a relatively stable initial deformation phase.
- **May 2025:** VH briefly rebounds to +4 mm, while VV remains negative (–7 mm), indicating a short-lived lateral recovery without vertical relief.
- **June 2025:** Both VH and VV return to negative values (–6 to –11 mm), confirming continued deformation activity with no clear stabilization trend toward the end of the monitoring window.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.896982856586153, 74.68214551819905)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement exhibits a generally positive and increasing trend, ranging roughly between –2 mm and +18 mm.
- **Vertical displacement (VV)** remains predominantly negative throughout the monitoring period, indicating overall vertical settlement at the A1 RHS location. VV values range approximately from –3 mm to –26 mm.

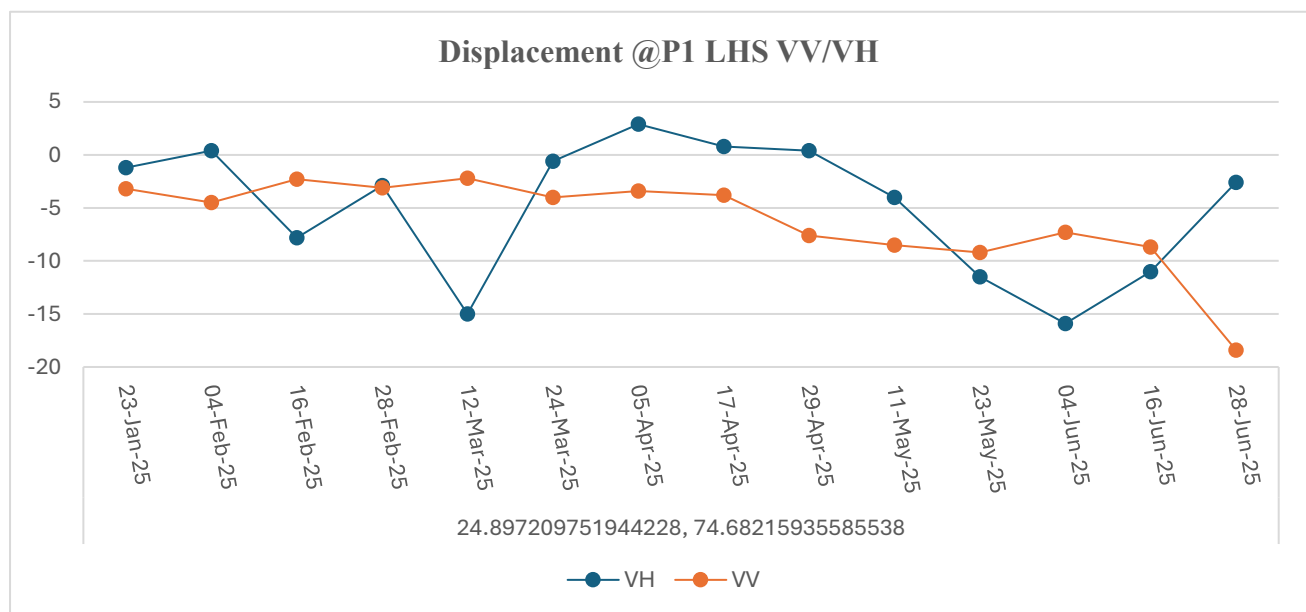
Critical Events & Anomalies:

- **January 2025:** VH begins at +7 mm, while VV is already negative (–4 mm), indicating early lateral displacement occurring alongside vertical settlement.
- **March 2025:** VH stabilizes around +5 mm, while VV improves slightly to –3 mm, indicating a temporary reduction in settlement intensity.
- **June 2025:** VH increases steadily to +18 mm, while VV remains strongly negative, confirming continued lateral displacement combined with unresolved vertical settlement toward the end of the monitoring period.

3.5 SAR Based Analysis at Pier 1 LHS:



P1 LHS (24.897209751944228, 74.68215935585538)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement shows a predominantly negative trend throughout the monitoring period (Jan–Jun 2025), indicating overall vertical settlement at the P1 LHS location.
- **Horizontal Displacement (VH)** Horizontal displacement exhibits moderate to high variability, ranging roughly between +3 mm and –16 mm.

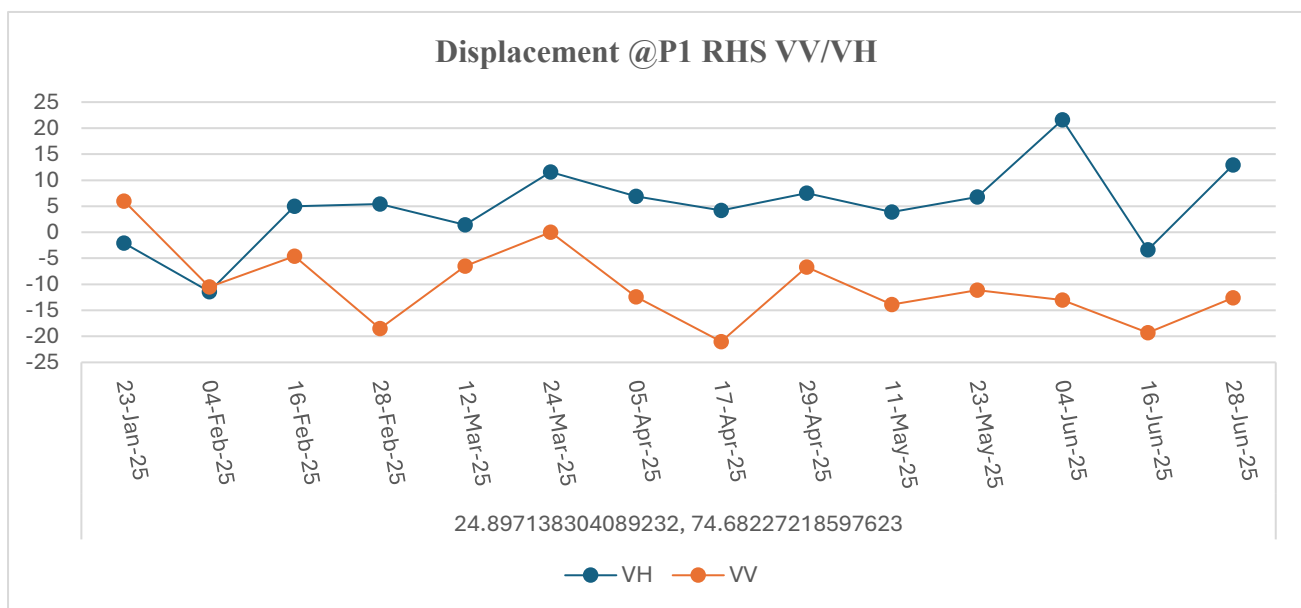
Critical Events & Anomalies:

- **January–February 2025:** VH remains near 0 to +1 mm, while VV is already negative (–3 to –5 mm), indicating early vertical settlement with minimal lateral response.
- **May 2025:** VH reaches –12 to –16 mm, while VV remains between –9 and –10 mm, marking a period of intensified lateral settlement with sustained vertical settlement.
- **June 2025:** VH recovers marginally to –3 mm, while VV drops to its most negative value (–18 mm), confirming severe vertical settlement persisting at the end of the monitoring period.

3.6 SAR Based Analysis at Pier 1 RHS:



P1 RHS (24.897138304089232, 74.68227218597623)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement remains predominantly negative throughout the monitoring period (Jan–Jun 2025), indicating overall vertical settlement at the P1 RHS location.
- **Horizontal Displacement (VH)** Horizontal displacement exhibits moderate to high variability with a predominantly positive trend, ranging roughly between –2 mm and +22 mm.

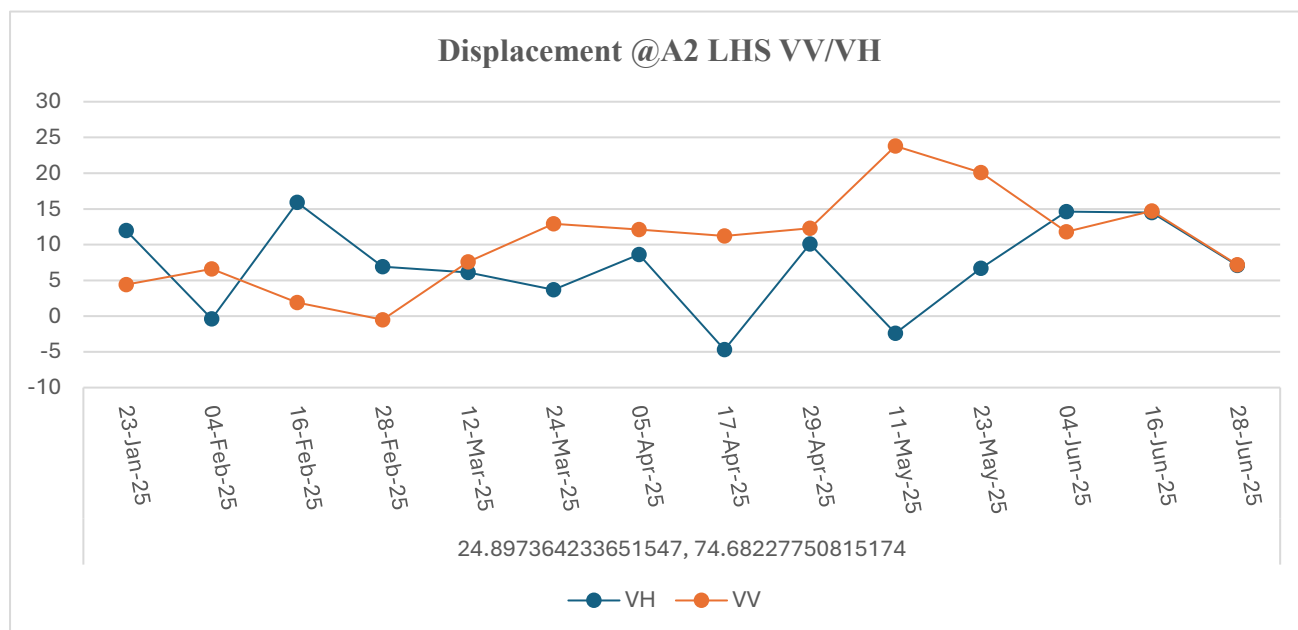
Critical Events & Anomalies:

- **January 2025:** VH begins near 0 mm, while VV is positive (+5 mm), indicating initial vertical uplift with minimal lateral displacement.
- **April 2025:** VV reaches one of its most negative values (–20 mm), while VH remains positive (+8 mm), representing a highly asymmetric deformation event.
- **June 2025:** VH drops briefly to 0 mm before recovering to +12 mm, while VV remains negative (–18 to –14 mm), confirming continued deformation with no vertical stabilization by the end of the monitoring period.

3.9 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.897364233651547, 74.68227750815174)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement exhibits moderate to high variability with a predominantly positive trend over the monitoring period (Jan–Jun 2025), indicating overall vertical uplift at the A2 LHS location.
- **Horizontal Displacement (VH)** Horizontal displacement shows high variability with alternating uplift and settlement phases, ranging roughly between +15 mm and –5 mm.

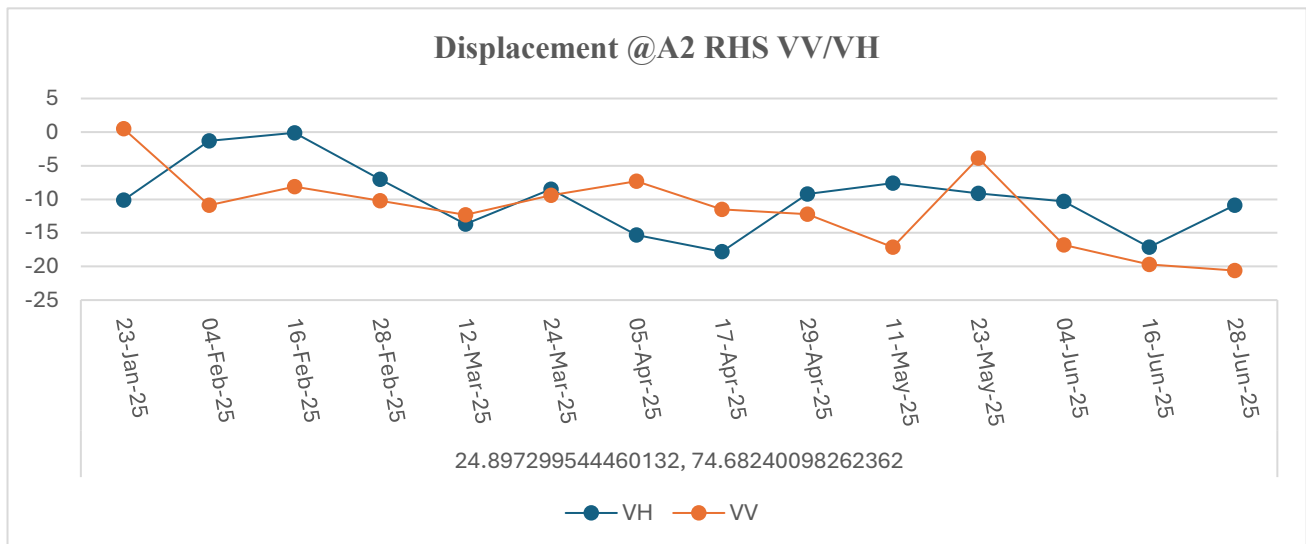
Critical Events & Anomalies:

- **February 2025:** VH reaches a local positive peak (+15 mm), while VV remains modest (+2–3 mm), indicating a lateral-dominant deformation phase with limited vertical participation.
- **June 2025:** VH recovers to +6–14 mm, while VV decreases to +12–14 mm, indicating post-peak vertical relaxation with renewed lateral displacement.

3.10 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.897299544460132, 74.68240098262362)



Key Observations:

Displacement Magnitude Patterns:

- Vertical Displacement (VV)** remains predominantly negative throughout the monitoring period, indicating overall vertical settlement at the A2 RHS location.
- Horizontal Displacement (VH)** Horizontal displacement exhibits moderate variability with a clear negative dominance, ranging roughly between -3 mm and -18 mm.

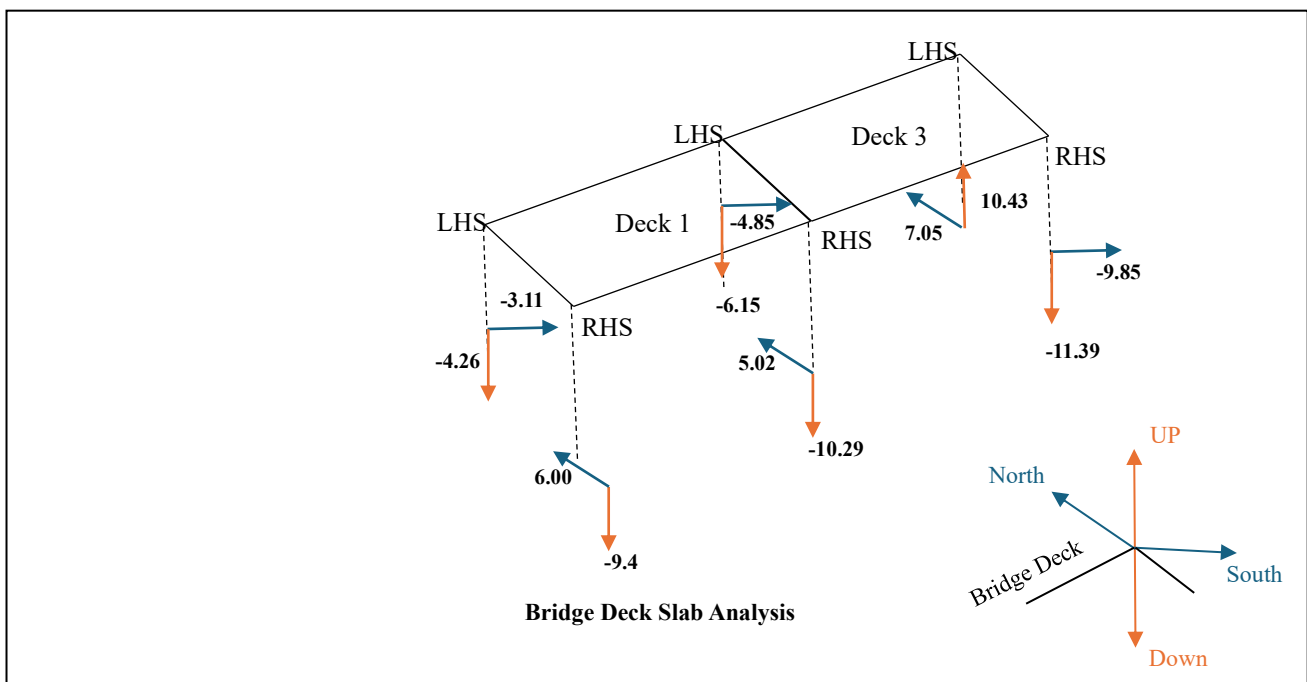
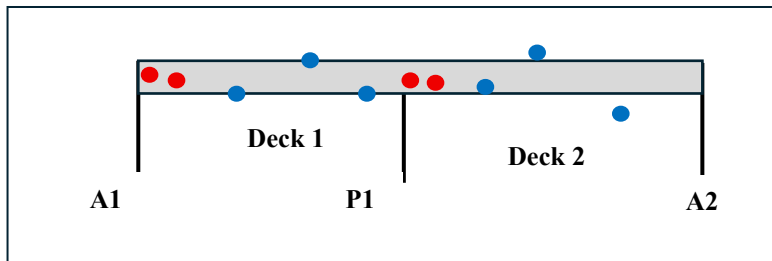
Critical Events & Anomalies:

- March 2025:** VH drops to -15 mm, while VV remains around -9 mm, highlighting a lateral-dominant settlement pulse.
- April 2025:** VV reaches one of its most negative values (-18 mm), while VH shows slight improvement (-9 mm), marking a severe vertical settlement event
- May-June 2025:** Both VH and VV deteriorate again (VH -18 mm, VV -20 mm), confirming renewed coupled settlement toward the end of the monitoring period.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The bridge deck slab exhibits significant non-uniform vertical displacement across Deck 1 to Deck 3, indicating pronounced differential vertical behaviour. Localized upward movement is limited and minor in comparison, confirming that vertical compression/settlement dominates the global deck response.

Horizontal Displacement Behaviour:

- Horizontal displacement shows clear directional variability and asymmetry across the deck, reflecting non-uniform lateral behaviour. The opposing horizontal displacement directions between LHS and RHS locations indicate torsional rotation rather than pure translational movement.

CONCLUSION:

- The bridge deck slab demonstrates a strongly non-symmetric deformation pattern, characterized by overall vertical settlement combined with differential horizontal movement. The concentration of maximum settlement and lateral displacement at the RHS of Deck 3 suggests localized stiffness variation, bearing response, or load eccentricity effects. The coexistence of widespread settlement and opposing horizontal displacements confirms that the deck behaviour is governed by torsional deformation and differential support response, rather than uniform vertical translation.